

17.4 - Normality in Random Events

Problem Solving, Math 1000

Bryce A Christopherson

1 Definition

A variable which, to a large extent, depends on the laws of chance is called a **random variable**. Think of this like the number of heads you get in some given amount of coin tosses – it varies with each trial; you wouldn't expect to get the same number of heads every single time.

2 Main Idea

The core notion here is that, while things might behave unusually on small scales (e.g. for some strange reason, you might get 97 heads in 100 flips every once in awhile), these random variables all have approximately normal distributions given a large enough number of observations. Indeed, it would be desirable to formalize some way to talk about this idea that we will approach normal distribution after some large enough number of trials. We do so by saying that, for n trials of our random variable, we would still get a bell shaped distribution, but the values of μ and σ might change. This leads to the following theorem:

3 Theorem (The Honest-Coin Principle)

Let X be a random variable representing the number of heads in n tosses of an honest (that is, fair) coin. Further, suppose that $n \geq 30$. Then, X has an approximately normal distribution with mean $\mu = n/2$ and standard deviation $\sigma = \sqrt{n}/2$.

4 Example

Suppose X is a random variable as above. For $n = 100$, we then get a mean of $\mu = 100/2 = 50$ and standard deviation of $\sigma = \sqrt{100}/2 = 5$.

However, we can generalize this idea further to even a dishonest coin (i.e. a 'trick' coin).

5 Theorem (Dishonest-Coin Principle)

Let X be a random variable representing the number of heads in n tosses of a dishonest coin with probability p of coming up heads. Further, assume $n \geq 30$. Then, X has an approximately normal distribution with mean $\mu = np$ and standard deviation $\sigma = \sqrt{np(1-p)}$.

6 Example

Suppose X is a random variable representing a dishonest coin as above. Suppose the coin has probability 0.2 (or 20%) of coming up heads. Then, if $n = 100$, we get a mean of $\mu = 100(0.2) = 20$ and standard deviation of $\sigma = \sqrt{100(0.2)(1-0.2)} = 4$.